Extending a marked group by a centralizing involution

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1 The extension problem

A string tuple of length n in a group G is an ordered tuple $(r_0, \dots, r_{n-1})$ of distinct nonidentity involutions such that $r_i r_j = r_j r_i$ whenever $|i - j| > 1$. It is a generating string marking when its entries generate G. No intersection property is required. The following argument isolates the group-theoretic step used in the half-turn constructions of Leemans–Toledo [1, Theorem 4.4 and Proposition 4.7], and in the corrected symmetric-end construction of the reviewed application [2, Sections 5–7].

The inputs from projective geometry remain separate. In the application, an even cyclic torus supplies an involution which centralizes an odd rotation dihedral subgroup. An exact maximal-subgroup theorem supplies the proper alternating or symmetric end. Neither existence statement is proved by the certificate below. Its conclusion concerns the actual four ordered generators, including their distinctness and generation.

Let (a, b, c) be a string tuple in G, and put

$$H = \langle a, b, c \rangle, \qquad D = \langle a, b \rangle.$$

We use $\mathrm{C}_H(a, b) = \{x \in H : xa = ax,\ xb = bx\}$; equivalently this is $H \cap \mathrm{C}_G(D)$. A maximal proper subgroup means an actual maximal member of the subgroup lattice below G, not merely a subgroup maximal within a selected list.

Lemma 1. *Suppose that z is a nonidentity involution commuting with a and b, and that z differs from a, b, c. Then (a, b, c, z) is a string tuple. Its first three entries generate exactly H.*

Proof. The old entries satisfy their original involution and distinctness conditions, and the hypotheses supply these conditions for the new entry. The nonconsecutive pairs of positions in a tuple of length four are $(0, 2), (0, 3), (1, 3)$. The corresponding commutations are $ac = ca$, $az = za$ and $bz = zb$, all of which are hypotheses. Restricting the new tuple to positions $0, 1, 2$ gives the original tuple entry by entry; therefore its generated subgroup is H. □

Lemma 2. *If $\mathrm{C}_H(a, b) = 1$, every nonidentity element z commuting with a and b lies outside H.*

Proof. If $z \in H$, the two commutations put it in $\mathrm{C}_H(a, b)$. The triviality assumption then gives $z = 1$, a contradiction. This argument does not require that z have finite order. □

Lemma 3. *If H is a maximal proper subgroup of G and $z \notin H$, then $\langle H, z \rangle = G$.*

Proof. The subgroup $\langle H, z \rangle$ contains H. Its containment is strict because it contains z, whereas H does not. Maximality leaves G as the only possibility. □

Theorem 4 (Full-end extension)**.** *Assume $H = G$, and let z be a nonidentity involution commuting with a, b. Either of the following additional conditions suffices to make (a, b, c, z) a generating string marking of G:*

1. *$ab \neq ba$ and $bc \neq cb$;*

2. *$z \notin D$ and $z \neq c$.*

The subgroup on its first three generators remains G.

Proof. Under the first condition, $z = a$ would imply $ab = ba$ by the commutation of z with b. The equality $z = b$ would similarly imply $ab = ba$ by commutation with a. Finally, $z = c$ would imply $bc = cb$. These three contradictions prove the required distinctness. Under the second condition, $a, b \in D$ immediately exclude the first two coincidences, and the last is explicitly excluded. Lemma 1 now supplies all string axioms. The new generated subgroup contains $H = G$, so it is G. The prefix is unchanged. □

Theorem 5 (Maximal-end extension)**.** *Assume that H is a maximal proper subgroup of G. Let z be a nonidentity involution commuting with a, b. If $z \notin H$, then (a, b, c, z) is a generating string marking of G, with first-three subgroup exactly H. In particular, the same conclusion follows from $\mathrm{C}_H(a, b) = 1$ without assuming $z \notin H$ separately.*

Proof. All three old generators belong to H. Thus $z \notin H$ makes z different from each of them. Lemma 1 proves the string axioms, and Lemma 3 proves generation. If the internal common centralizer is trivial, Lemma 2 first proves the required nonmembership. Prefix equality again identifies the actual old subgroup; this conclusion is stronger than an abstract isomorphism of unnamed subgroups. □

2 Relation to the projective-line application

For the full-end case in [2, Section 5], let q be odd, choose $\alpha \in \{1, -1\}$ with $q \equiv \alpha \pmod 4$, and put $N = (q - \alpha)/2$. The application retains the usual split and nonsplit tori of $\mathrm{PSL}_2(q)$ and their normalizers. If the first adjacent order is an odd divisor $k > 1$ of N, the relevant torus has order N, which is even. Its unique involution z centralizes the normalizer and hence D. Since D has odd rotation order and trivial centre, $z \notin D$. The next adjacent pair is noncommuting, so $z \neq c$. Theorem 4(2) is the final extension step. Alternatively, Theorem 4(1) uses the already proved noncommutation of both adjacent pairs and avoids repeating the distinctness argument.

For the proper-end cases, the accepted application proves $\mathrm{C}_H(a, b) = 1$ for its chosen D_{10} in A_5 and S_3 in S_4. It retains Giudici's exact maximal-subgroup theorem in the applicable prime ranges [3, Theorem 2.2(6),(8)]. The torus still supplies z, and Theorem 5 then proves generation and the actual facet subgroup. The following steps remain in the written application: the opposite end is full by the classification; the marked-group action produces a regular maniplex; subgroup indices give the face counts; and simplicity gives nonorientability. None is silently folded into the new Lean conclusion.

3 Formal interface and limitations

The Lean file defines local string tuples and generating string markings using an actual function $\mathrm{Fin}(n) \to G$. Its conclusions contain nonidentity, involution equations, function injectivity, every distant commutation and subgroup-closure equality with the whole group. The concrete tuple is $[a, b, c, z]$. The theorem `prefix_preserved` identifies its restriction with $[a, b, c]$. The internal-centralizer hypothesis is the subgroup equality

$$H \cap \mathrm{C}_G(\{a, b\}) = \{1\},$$

represented by an infimum and the bottom subgroup. Maximality is `IsCoatom H` in the full subgroup lattice.

The Lean certificate proves the lemmas and extension theorems above for arbitrary groups. It assumes neither the four-generator conclusion nor the desired final generation statement. It does assume the stated old triple, and the projective applications must supply its hypotheses. There is no formalization here of projective groups, torus existence, Dickson or Giudici classification, the finite local census, the intersection property, maniplexes, or the final infinite-family statements. The compiled declarations use only Lean's standard logical axioms. Library dependencies and their exact revisions are recorded in the replay package; the imported compiled library remains trusted.

An elementary finite control explains why the outside condition is substantive even if distinctness is independently known. In the additive group $G = \mathbb{F}_2^4$, write products additively and take $a = e_1$, $b = e_2$, $c = e_3$, $z = e_1 + e_2 + e_3$. The subgroup $H = \langle a, b, c \rangle$ is a maximal hyperplane. All four entries are distinct nonzero involutions and all commute, but they still generate H, not G. Here $z \in H$ and $\mathrm{C}_H(a, b) = H$ is nontrivial. The companion finite checker exhausts the sixteen elements and verifies this countermodel, including maximality by testing every outside adjoining element. It is an independent finite check of the weakened statement's failure, not the proof of the general theorems.

References

[1] D. Leemans and M. Toledo, *Maniplexes with automorphism group* $\mathrm{PSL}(2, q)$, author manuscript dated 15 May 2023, eleven pages, Theorem 4.4 and Proposition 4.7.

[2] human_compression project, *Centralizer extensions of projective-line maniplexes*, reviewed phase258 application, Sections 5–7; independent review phase263.

[3] M. Giudici, *Maximal subgroups of almost simple groups with socle* $\mathrm{PSL}(2, q)$, arXiv:math/0703685v1, 2007, Theorem 2.2.